



FACULTY OF AGRICULTURE BHABHA UNIVERSITY

Jatkhedi, NH-12 , Hoshangabad Road , Bhopal,
Madhya Pradesh- 462026

Bachelor of Science in Agriculture SEMESTER-WISE DISTRIBUTION OF COURSES

SCHEME

SEMESTER-III			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Crop Production Technology – I (<i>Kharif Crops</i>)	BAG-5301	2 (1+1)
2.	Fundamentals of Plant Breeding	BAG-5302	3(2+1)
3.	Agricultural Finance and Cooperation	BAG-5303	3(2+1)
4.	Agri- Informatics	BAG-5304	2 (1+1)
5.	Farm Machinery and Power	BAG-5305	2 (1+1)
6.	Production Technology for Vegetables and Spices	BAG-5306	2 (1+1)
7.	Environmental Studies and Disaster Management	BAG-5307	3(2+1)
8.	Statistical Methods	BAG-5308	2 (1+1)
9.	Livestock and Poultry Management	BAG-5309	4 (3+1)
TOTAL			23(14+9)

SEMESTER-IV			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Crop Production Technology –II (<i>Rabi Crops</i>)	BAG-5401	2 (1+1)
2.	Production Technology for Ornamental Crops, MAP and Landscaping	BAG-5402	2 (1+1)
3.	Renewable Energy and Green Technology	BAG-5403	2 (1+1)
4.	Problematic Soils and their Management	BAG-5404	2 (2+0))
5.	Production Technology for Fruit and Plantation Crops	BAG-5405	2 (1+1)
6.	Principles of Seed Technology	BAG-5406	3(2+1)
7.	Farming System & Sustainable Agriculture	BAG-5407	1(1+0)
8.	Agricultural Marketing Trade & Prices	BAG-5408	3(2+1)
9.	Introductory Agro-meteorology & Climate Change	BAG-5409	2(1+1)
10.	Elective Course	BAG-5410	3 credit
TOTAL			19(11+8) + 3



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SYLLABUS

SECOND YEAR

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Crop Production Technology – I (<i>Kharif Crops</i>)	BAG-5301	2(1+1)

THEORY

Unit I

Origin, geographical distribution, economic importance, soil and rice, pigeonpea, soybean

Unit II

Origin, geographical distribution, economic importance, soil and maize, mungbean and groundnut, jute

Unit III

Origin, geographical distribution, economic importance, soil and sorghum, urdbean, cowpea, cotton

Unit IV

Origin, geographical distribution, economic importance, soil and pearl millet, forage crops-sorghum

Unit V

Origin, geographical distribution, economic importance, soil and finger millet, cluster bean and napier

PRACTICAL

Rice nursery preparation, transplanting of rice, sowing of soybean, pigeonpea and mungbean. maize, groundnut and cotton, effect of seed size on germination and seedling vigour of kharif season crops, effect of sowing depth on germination of kharif crops, identification of weeds in kharif season crops, top dressing and foliar feeding of nutrients, study of yield contributing characters and yield calculation of kharif season crops, study of crop varieties and important agronomic experiments at experimental farm. study of forage experiments, morphological description of kharif season crops, visit to research centres of related crops.



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SECOND YEAR

SEMESTER

III		
SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Plant Breeding	BAG-5302	3(2+1)

THEORY

Unit-I

Historical development, concept, nature, role, aim and objectives of plant breeding. Major achievements and future prospects.

Domestication, Acclimatization and Introduction. Centre of origin/diversity.

Modes of reproduction and apomixes and its genetic consequences.

Unit-II

Concepts of population genetics and Hardy-Weinberg Law.

Self-incompatibility and male-sterility.

Genetics in relation to plant breeding; Components of Genetic variation. Heritability and genetic advance.

Unit-III

Genetic basis and breeding methods in self pollinated crops- mass selection and pure line selection, hybridization techniques and handling of segregating population. Multiline concept.

Unit-IV

Heterosis and inbreeding depression. Development of inbred lines and hybrids, composite and synthetic varieties. Genetic basis and methods of breeding for cross pollinated crops, modes of selection. Population improvement Schemes- Ear to row method, Modified Ear to Row method, Recurrent selection schemes.

Unit-V

Breeding methods in asexually propagated crops- clonal selection and hybridization. Wide hybridization and pre-breeding. Polyploidy in relation to plant breeding. Mutation breeding- methods and uses. Breeding for important biotic and abiotic stresses. Biotechnological tools- DNA markers and marker assisted selection. Participatory plant breeding; Intellectual Property Rights, Patenting, Plant



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Breeders and & Farmer's Rights. Maintenance of breeding records and data collection.

PRACTICAL

- Plant Breeder's kit.
- Study of germplasm of various crops.
- Study of floral structure of self-pollinated and cross pollinated crops.
- Emasculation and hybridization techniques in self & cross pollinated crops.
- Consequences of inbreeding on genetic structure of resulting populations.
- Study of male sterility system.
- Handling of segregation populations.
- Methods of calculating mean, range, variance, standard deviation and heritability.
- Designs used in plant breeding experiment; analysis of Randomized Block Design
- Estimation of heterosis and inbreeding depression.
- Layout of field experiments; Work out the mode of pollination in a given crop and extent of natural out crossing.
- Prediction of performance of double cross hybrids.



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SECOND YEAR

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agricultural Finance and Co-Operation	BAG-5303	3(2+1)

THEORY

UNIT I

Agricultural Finance- meaning, scope and significance, credit needs and its role in Indian agriculture. Agricultural credit: meaning, definition, need, classification. Credit analysis: 4 R's, and 3C's of credits.

UNIT II

Sources of agricultural finance: institutional and non-institutional sources, commercial banks, social control and nationalization of commercial banks, Micro financing including KCC. Lead bank scheme, RRBs, Scale of finance and unit cost. An introduction to higher financing institutions – RBI, NABARD, ADB, IMF, world bank, Insurance and Credit Guarantee Corporation of India. Cost of credit. Recent development in agricultural credit.

UNIT III

Preparation and analysis of financial statements – Balance Sheet and Income Statement. Basic guidelines for preparation of project reports- Bank norms – SWOT analysis.

UNIT IV

Agricultural Cooperation – Meaning, brief history of cooperative development in India, objectives, principles of cooperation, significance of cooperatives in Indian agriculture.

UNIT V

Agricultural Cooperation in India- credit, marketing, consumer and multi-purpose cooperatives, farmers' service cooperative societies, processing cooperatives, farming cooperatives, cooperative warehousing; role of ICA, NCUI, NCD, NAFED.

PRACTICAL

Determination of most profitable level of capital use. Optimum allocation of limited amount of capital among different enterprise. Analysis of progress and performance of cooperatives using published data. Analysis of progress and performance of commercial banks and RRBs using published data. Visit to a commercial bank, cooperative bank and cooperative society to acquire firsthand knowledge of their management, schemes and procedures. Estimation of credit



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requirement of farm business – A case study. Preparation and analysis of balance sheet – A case study. Preparation and analysis of income statement – A case study. Appraisal of a loan proposal – A case study. Techno-economic parameters for preparation of projects. Preparation of Bankable projects for various agricultural products and its value added products. Seminar on selected topics.

SECOND YEAR

SYLLABUS

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agri-Informatics	BAG-5304	2(1+1)

THEORY

Unit I

Introduction to Computers, Operating Systems, definition and types, Applications of MSOffice for document creation & Editing, Data presentation, interpretation and graph creation, statistical analysis, mathematical expressions,

Unit II

Database, concepts and types, uses of DBMS in Agriculture, World Wide Web (WWW): Concepts and components. Introduction to computer programming languages, concepts and standard input/output operations.

Unit III

e-Agriculture, concepts and applications, Use of ICT in Agriculture. Computer Models for understanding plant processes.

Unit IV

IT application for computation of water and nutrient requirement of crops, Computer-controlled devices (automated systems) for Agri-input management, Smartphone Apps in Agriculture for farm advises, market price, postharvest management etc;

Unit V

Geospatial technology for generating valuable agri-information. Decision support systems, concepts, components and applications in Agriculture, Agriculture Expert System, Soil Information Systems etc for supporting Farm decisions. Preparation of contingent crop-planning using IT tools.



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PRACTICAL

Study of Computer Components, accessories, practice of important DOS Commands. Introduction of different operating systems such as windows, Unix/ Linux, Creating, Files & Folders, File Management. Use of MS-WORD and MS Power-point for creating, editing and presenting a scientific Document. MS- EXCEL - Creating a spreadsheet, use of statistical tools, writing expressions, creating graphs, analysis of scientific data. MS- ACCESS: Creating Database, preparing queries and reports, demonstration of Agri-information system. Introduction to World Wide Web (WWW). Introduction of programming languages. Hands on Crop Simulation Models (CSM) such as DSSAT/Crop- Info/CropSyst/Wofost; Computation of water and nutrient requirements of crop using CSM and IT tools. Introduction of Geospatial Technology for generating valuable information for Agriculture. Hands on Decision Support System. Preparation of contingent crop planning.

SECOND YEAR

SYLLABUS

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Farm Machinery and Power	BAG-5305	2(1+1)

THEORY

Unit I

Status of Farm Power in India, Sources of Farm Power

Unit II

I.C. engines, working principles of I C engines, comparison of two stroke and four stroke cycle engines, Study of different components of I.C. engine, I.C. engine terminology and solved problems

Unit III

Familiarization with different systems of I.C. engines: Air cleaning, cooling, lubrication, fuel supply and hydraulic control system of a tractor

Unit IV

Familiarization with Power transmission system: clutch, gear box, differential and final drive of a tractor, Tractor types, Cost analysis of tractor power and attached implement,

Unit V

Familiarization with Primary and Secondary Tillage implement, Implement for hill



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agriculture, implement for intercultural operations, Familiarization with sowing and planting equipment, calibration of a seed drill and solved examples, Familiarization with Plant Protection equipment, Familiarization with harvesting and threshing equipment.

PRACTICAL

Study of different components of I.C. engine. To study air cleaning and cooling system of engine, Familiarization with clutch, transmission, differential and final drive of a tractor, Familiarization with lubrication and fuel supply system of engine, Familiarization with brake, steering, hydraulic control system of engine, Learning of tractor driving, Familiarization with operation of power tiller, Implements for hill agriculture, Familiarization with different types of primary and secondary tillage implements: mould plough, disc plough and disc harrow . Familiarization with seed cum-fertilizer drills their seed metering mechanism and calibration, planters and transplanter Familiarization with different types of sprayers and dusters Familiarization with different intercultural equipment, Familiarization with harvesting and threshing machinery.

SYLLABUS

SECOND YEAR

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Production Technology for Vegetables and Spices	BAG-5306	2(1+1)

THEORY

Unit I

Importance of vegetables & spices in human nutrition and national economy, kitchen gardening. Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorders, of important vegetable and spices

: Tomato, Brinjal, Chilli, Capsicum.

Unit II

Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorders : Cucumber, Melons, Gourds, Pumpkin,

Unit III



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Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorders : French bean, Peas;

Unit IV

Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorder: Cole crops such as Cabbage, Cauliflower, Knol-khol

Unit V

Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorder: Bulb crops such as Onion, Garlic; Rootcrops such as Carrot, Radish, Beetroot; Tuber crops such as Potato; Brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorder: Leafy vegetables such as Amaranth, Palak. Perennial vegetables.

PRACTICAL

Identification of vegetables & spice crops and their seeds. Nursery raising. Direct seed sowing and transplanting. Study of morphological characters of different vegetables & spices. Fertilizers applications. Harvesting & preparation for market. Economics of vegetables and spices cultivation.

SECOND YEAR

SYLLABUS

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Environmental Studies and Disaster Management	BAG-5307	3(2+1)

THEORY

UNIT I

Multidisciplinary nature of environmental studies Definition, scope and importance. Natural Resources: Renewable and non-renewable resources, Natural resources and associated problems. a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people. b) Water resources: Use and over-utilization of



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surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. e) Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources. Case studies. f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification. Role of an individual in conservation of natural resources. Equitable use of resources for sustainable life styles.

UNIT II

Environmental Pollution: definition, cause, effects and control measures of: a. Air pollution b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation. Public awareness.

UNIT III

Ecosystems: Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem. Ecological succession, Food chains, food webs and ecological pyramids. Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries). Biodiversity and its conservation: - Introduction, definition, genetic, species & ecosystem diversity and biogeographical classification of India. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. Biodiversity at global, National and local levels, India as a mega-diversity nation. Hot-spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India. Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT IV

Social Issues and the Environment: From Unsustainable to Sustainable development, Urban problems related to energy, Water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Wasteland reclamation. Consumerism and waste products. Human Population and the Environment: population growth, variation among nations, population explosion, Family Welfare Programme. Environment and human health: Human Rights, Value Education, HIV/AIDS. Women and Child Welfare. Role of Information Technology in



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Environment and human health.

UNIT V

Natural Disasters- Meaning and nature of natural disasters, their types and effects. Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves, Climatic change: global warming, Sea level rise, ozone depletion. Man Made Disasters- Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, air pollution, water pollution, deforestation, industrial waste water pollution, road accidents, rail accidents, air accidents, sea accidents. Disaster Management- Effect to migrate natural disaster at national and global levels. International strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community –based organizations and media. Central, state, district and local administration; Armed forces in disaster response; Disaster response; Police and other organizations.

PRACTICAL

- Pollution case studies
- Case Studies- Field work: Visit to a local area to document environmental assets river/ forest/ grassland/ hill/ mountain,
- Visit to a local polluted site-Urban/Rural/Industrial/ Agricultural
- Study of common plants, insects, birds
- Study of simple ecosystems-pond, river, hill slopes, etc.

SYLLABUS

SECOND YEAR

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Statistical Methods	BAG-5308	2(1+1)

THEORY

Unit I

Introduction to Statistics and its Applications in Agriculture, Graphical Representation of Data, Measures of Central Tendency & Dispersion.

Unit II

Definition of Probability, Addition and Multiplication Theorem (without proof). Simple Problems Based on Probability. Binomial & Poisson Distributions.



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Unit III

Definition of Correlation, Scatter Diagram.Karl Pearson's Coefficient of Correlation.Linear Regression Equations.

Unit IV

Introduction to Test of Significance, One sample & two sample test t for Means, Chi-Square Test of Independence of Attributes in 2 ×2 Contingency Table.

Unit V

Introductions to Analysis of Variance, Analysis of One Way Classification. Introduction to Sampling Methods, Sampling versus Complete Enumeration, Simple Random Sampling with and without replacement, Use of Random Number Tables for selection of Simple Random Sample.

PRACTICAL

Graphical Representation of Data.Measures of Central Tendency (Ungrouped data) with Calculation of Quartiles, Deciles& Percentiles.Measures of Central Tendency (Grouped data) with Calculation of Quartiles, Deciles& Percentiles.Measures of Dispersion (Ungrouped Data).Measures of Dispersion (Grouped Data).Moments, Measures of Skewness& Kurtosis (Ungrouped Data).Moments, Measures of Skewness& Kurtosis (Grouped Data).Correlation & Regression Analysis.Application of One Sample t-test.Application of Two Sample Fisher's t-test.Chi-Square test of Goodness of Fit.Chi-Square test of Independence of Attributes for 2 ×2 contingency table.Analysis of Variance One Way Classification.Analysis of Variance Two Way Classification.Selection of random sample using Simple Random Sampling.

SECOND YEAR

SYLLABUS

SEMESTER III

SUBJECT TITLE	SUBJECT CODE	CREDIT
Livestock & Poultry Management	BAG-5309	4 (3+1)

THEORY

UNIT I

Role of livestock in the national in species farm animals and of livestock poultry.Housingprinciple economy.Reproduction space and poultry.Management of calves,growing calves and milked animals.



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UNIT II

Management of sheep, goat and swine. Incubation, hatching and brooding. Management of growers and layers.

UNIT III

Important Indian and exotic breeds of cattle, buffalo, sheep, goat, swine and poultry. Improvement of farm animals and poultry.

UNIT IV

Digestion in livestock and poultry. Classification of feedstuffs. Proximate principles of feed. Nutrients and their functions. Feed ingredients for ration for livestock and poultry. Feed supplements and feed additives.

UNIT V

Feeding of livestock and poultry. Introduction of livestock and poultry diseases. Prevention (including vaccination schedule) and control of important diseases of livestock and poultry.

PRACTICAL

External body parts of cattle, buffalo, sheep, goat, swine and poultry. Handling and restraining of livestock. Identification methods of farm animals and poultry. Visit to IDF and IPF to study breeds of livestock and poultry and daily routine farm operations and farm records. Judging of cattle, buffalo and poultry. Culling of livestock and poultry. Planning and layout of housing for different types of livestock. Computation of rations for livestock. Formulation of concentrate mixtures. Clean milk production, milking methods. Hatchery operations, incubation and hatching equipments. Management of chicks, growers and layers. Debeaking, dusting and vaccination. Economics of cattle, buffalo, sheep, goat, swine and poultry production.



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Bachelor of Science in Agriculture SYLLABUS

SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Crop Production Technology –II (Rabi Crops)	BAG-5401	2(1+1)

THEORY

Unit I

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of Rabi crops; wheat and chickpea, mustard, berseem.

Unit II

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of Barley, lentil, peas, sugarcane.

Unit III

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of Rapeseed, sunflower and oat.

Unit IV

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of lucerne and Citronella

Unit V

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of Mentha and Lemon grass

PRACTICAL

Sowing methods of wheat and sugarcane, identification of weeds in *rabiseason* crops, study of morphological characteristics of *rabicrops*, study of yield contributing characters of *rabiseason* crops, yield and juice quality analysis of sugarcane, study of important agronomic experiments of *rabicrops* at experimental farms. Study of *rabiforage* experiments, oil extraction of medicinal crops, visit to research stations of related crops.



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SECOND YEAR SYLLABUS SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Production Technology for Ornamental Crops, MAPs and Landscaping	BAG-5402	2(1+1)

THEORY

Unit I

Importance and scope of ornamental crops, medicinal and aromatic plants landscaping. Principles of landscaping. Landscape uses of trees, shrubs and climbers.

Unit II

Production technology of important cut flowers like rose, gerbera, carnation, liliun and orchids under protected conditions and gladiolus, tuberose, chrysanthemum under open conditions.

Unit III

Package of practices for looseflowers like marigold and jasmine under open conditions.

Unit IV

Production technology of important medicinal plants like ashwagandha, asparagus, aloe, costus, Cinnamomum, periwinkle, isabgol and aromatic plants like mint, lemongrass, citronella, palmarosa, ocimum, rose, geranium, vetiver.

Unit V

Processing and value addition in ornamental crops and MAPs produce.

PRACTICAL

Identification of Ornamental plants. Identification of Medicinal and Aromatic Plants. Nursery garden preparation and seed sowing. Training and mapping of Ornamental plants. Planning and layout of landscape. Intercultural operations in flowers and MAP. Harvesting and post harvest handling of cut and loose flowers. Processing of MAP. Visit to commercial flower/MAP unit.



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SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Renewable Energy and Green Technology	BAG-5403	2(1+1)

THEORY

Unit I

Classification of energy sources, contribution of these of sources in agricultural sector

Unit II

Familiarization with biomass utilization for biofuel production and their application, Familiarization with types of biogas plants and gasifiers, biogas, bioalcohol, biodiesel and biooil production and their utilization as bioenergy resource

Unit III

Introduction of solar energy, collection and their application

Unit IV

Familiarization with solar energy gadgets: solar cooker, solar water heater, application of solar energy: solar drying, solar pond, solar distillation, solar photovoltaic system and their application,

Unit V

Introduction of wind energy and their application.

PRACTICAL

Familiarization with renewable energy gadgets. To study biogas plants, To study gasifier, To study the production process of biodiesel, To study briquetting machine, To study the production process

of bio-fuels. Familiarization with different solar energy gadgets. To study solar photovoltaic system: solar light, solar pumping, and solar fencing. To study solar cooker, To study solar drying system. To study solar distillation and solar pond.



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Bachelor of Science in Agriculture SYLLABUS

SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Problematic Soils and their Management	BAG-5404	2(2+0)

THEORY

UNIT I

Soil quality and health, Distribution of Waste land and problem soils in India. Their categorization based on properties.

UNIT II

Irrigation water – quality and standards, utilization of saline water in agriculture.

UNIT III

Reclamation and management of Saline and sodic soils, Acid soils, Acid Sulphate soils, Eroded and Compacted soils, Flooded soils, Polluted soils.

UNIT IV

Remote sensing and GIS in diagnosis and management of problem soils. Problematic soils under different Agro-ecosystems.

UNIT V

Multipurpose tree species, bio remediation through MPTs of soils, land capability and classification, land suitability classification.

SYLLABUS

SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Production Technology for Fruit and Plantation Crops	BAG-5405	2(1+1)



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THEORY

UNIT I

Importance and scope of fruit and plantation crop industry in India; Importance of rootstocks; Production technologies for the cultivation of major fruits-mango, banana,

UNIT II

Production technology for citrus, grape, guava, litchi, papaya.

UNIT III

Production technology for sapota, apple, pear, peach, walnut, almond.

UNIT IV

Production technology for minor fruits- date, ber, pineapple, pomegranate, jackfruit, strawberry,

UNIT V

Production technology for plantation crops- coconut, arecanut, cashew, tea, coffee & rubber.

PRACTICAL

Seed propagation. Scarification and stratification of seeds. Propagation methods for fruit and plantation crops. Description and identification of fruit. Preparation of plant bio regulators and their uses, Important pests, diseases and physiological disorders of above fruit and plantation crops, Visit to commercial orchards.

SYLLABUS

SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Principles of Seed Technology	BAG-5406	3(2+1)

THEORY

Unit I

Difference between seed and grain, Quality seed and its importance in Agriculture, Core Concept of seed quality, Seed technology; definitions, objectives and its role in increasing agricultural



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production. Seed improvement programmes in India and Madhya Pradesh. Generation system of seed production- Nucleus seed breeder seed, foundation seed, certified seed and truthfully leveled seed.

Unit II

Reproduction process in crop plants; sexual and asexual reproduction, apomixes. Seed formation and development. Seed replacement and multiplication rates. Seed demand forecasting. Principles of seed production. Maintenance of genetic purity causes of varietal deterioration and its control.

Unit III

Male sterility- concepts and its use in hybrid seed production; inbred and non-inbred lines. Nucleus and Breeder seed production of newly released and established varieties of self pollinated crops like rice, wheat, soybean, chickpea, pigeon pea, rapeseed, mustard etc. Foundation and Certified seed production of maize hybrids, single and double cross hybrids. Hybrid seed production of rice, sunflower, sorghum and pearl millet using male sterility system. Latest released hybrids, their parentage and characteristics. Foundation and Certified seed production of wheat, rice, soybean, grams, sunflower, pigeon pea, groundnut, castor, cotton etc. Foundation and Certified seed production of some important vegetables like onion, brinjal, chillies, tomato, okra and gourds etc.

Unit IV

Seed Certification- its concepts, role, goals and procedure. Seed certification agencies. Minimum certification standards for self and cross pollinated crops. Field and seed inspection, its objectives. Seed Act and Seed Rules; Seed Legislation and Seed Law Enforcement, Seed Control Orders, Seed Policies, Seed Bills, WTO, IPR, PBR in India and recent development in Indian Seed Industry.

Unit V

Seed quality regulations; seed processing-cleaning, grading seed treatment methods, bagging and storage, factors affecting seed quality in storage, storage pests and disease control. Orthodox and recalcitrant seeds. Seed testing- principles and methods of sampling, purity analysis, seed moisture, germination, viability and vigor test. Cultivar purity testing- ODV, electrophoresis and grow-out tests for seed genetic purity, seed health etc. Seed dormancy; types, causes and breaking methods. Seed marketing, organizations, seed pricing, promotion of quality seeds and seed marketing strategies etc.



FACULTY OF AGRICULTURE

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Madhya Pradesh- 46202

Bachelor of Science in Agriculture

PRACTICAL

- Study of reproductive systems in crop plants-floral biology, pollination and fertilization.
- Classification of seeds based on their usage, preparation of seed album.
- Study of seed structure in monocots and dicots.
- Study of seed production in major crops- MS lines, pollen shedders, off-types, emasculation and pollination techniques.
- Study of seed processing equipments- plan and layout.
- Study of seed testing equipments, seed testing methods- seed sampling, seed purity test, seed moisture, seed germination tests, seed viability and vigor, seed health tests etc. s
- Seed dormancy breaking methods.
- Visit to seed production plots
- Visit to seed processing plants, public and private seed enterprises.

SECOND YEAR

SYLLABUS

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Farming System & Sustainable Agriculture	BAG-5407	1(1+0)

THEORY

Unit I

Farming System-scope, importance, and concept, Types and systems of farming system and factors affecting types of farming, Farming system components and their maintenance.

Unit II

Cropping system and pattern, multiple cropping system, Efficient cropping system and their evaluation, Allied enterprises and their importance, Tools for determining production and efficiencies in cropping and farming system.

Unit III

Sustainable agriculture-problems and its impact on agriculture, indicators of sustainability, adaptation and mitigation, conservation agriculture strategies in agriculture, HEIA, LEIA and LEISA and its techniques for sustainability, Integrated farming system-historical background, objectives



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and characteristics, components of IFS and its advantages.

Unit IV

Site specific development of IFS model for different agro-climatic zones, resource use efficiency and optimization techniques, Resource cycling and flow of energy in different farming system, farming system and environment.

Unit V

Visit of IFS model in different agro-climatic zones of nearby states University/ institutes and farmers field.

SECOND YEAR

SYLLABUS

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agricultural Marketing Trade & Prices	BAG-5408	3(2+1)

THEORY

Unit I

Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing, market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets; demand, supply and producer's surplus of agri-commodities: nature and determinants of demand and supply of farm products,

Unit II

producer's surplus – meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agri-commodities; product life cycle (PLC) and competitive strategies: Meaning and stages in PLC; characteristics of PLC; strategies in different stages of PLC; pricing and promotion strategies: pricing considerations and approaches – cost based and competition based pricing; market promotion – advertising, personal selling, sales promotion and publicity – their meaning and merits & demerits; marketing process and functions:

Unit III

Marketing process-concentration, dispersion and equalization; exchange functions – buying and selling; physical functions – storage, transport and processing; facilitating functions – packaging, branding, grading, quality control and labeling (Agmark); Market functionaries and marketing channels: Types and importance of agencies involved in agricultural marketing;



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Unit IV

meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products; Integration, efficiency, costs and price spread: Meaning, definition and types of market integration; marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing; reasons for higher marketing costs of farm commodities; ways of reducing marketing costs; Role of Govt. in agricultural marketing:

Unit V

Public sector institutions- CWC, SWC, FCI, CACP & DMI – their objectives and functions; cooperative marketing in India; Risk in marketing: Types of risk in marketing; speculation & hedging; an overview of futures trading; Agricultural prices and policy: Meaning and functions of price; administered prices; need for agricultural price policy; Trade: Concept of International Trade and its need, theories of absolute and comparative advantage. Present status and prospects of international trade in agri- commodities; GATT and WTO; Agreement on Agriculture (AoA) and its implications on Indian agriculture; IPR.

PRACTICAL

Plotting and study of demand and supply curves and calculation of elasticities; Study of relationship between market arrivals and prices of some selected commodities; Computation of marketable and marketed surplus of important commodities; Study of price behaviour over time for some selected commodities; Construction of index numbers; Visit to a local market to study various marketing functions performed by different agencies, identification of marketing channels for selected commodity, collection of data regarding marketing costs, margins and price spread and presentation of report in the class; Visit to market institutions – NAFED, SWC, CWC, cooperative marketing society, etc. to study their organization and functioning; Application of principles of comparative advantage of international trade.



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SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Introductory Agro-meteorology & Climate Change	BAG-5409	2(1+1)

THEORY

Unit I

Agricultural Meteorology- Introduction, definition of meteorology, scope and practical utility of Agricultural meteorology. Composition and structure of atmosphere and definition of weather and climate, aspects involved in weather and climate, atmospheric temperature, soil temperature, solar radiation, atmospheric pressure, atmospheric humidity, evaporation and transpiration, monsoons, rainfall, clouds, drought, weather disasters and their management atmospheric pollution and role of meteorology.

Unit II

Basics of weather forecasting. Climate change-causes. Global warming-causes and remote sensing. Effect of climate change on horticulture. Past and future changes in greenhouse gases within the atmosphere. Sources and sinks for greenhouse gases. Atmospheric chemistry. Plants sense and respond to changes in CO₂ concentration.

Unit III

Measurement of short-term effects and mechanisms underlying the observed responses in C₃ and C₄ species. Plant development affected by growth in elevated CO₂. Physiology of rising CO₂ on nitrogen use and soil fertility, its implication for production. Methodology for studying effect of CO₂. Change in secondary metabolites and pest disease reaction of plants.

Unit IV

The mechanisms of ozone and UV damage and tolerance in plants. Increased temperature and plants in tropical/sub-tropical climates- effect on growing season, timing of flowering, duration of fruit development and impacts on crop yields and potential species ranges, interaction of temperature with other abiotic/biotic stress. Mitigation strategies and prospects for genetic manipulation of crops to maximize production in the future atmosphere.

Unit V

Modifying Rubisco, acclimation, metabolism of oxidizing radicals, and sink capacity as potential strategies.



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PRACTICAL

Site selection for Agromet observatory; Measurement of temperature; Measurement of rainfall;

Measurement of evaporation (atmospheric/soil); Measurement of atmospheric pressure; Measurement of sunshine duration and solar radiation; Measurement of wind direction and speed and relative humidity; Study of weather forecasting and synoptic charts. Visit to Meteorological observatory, Visit to IMD meteorological observatory-Lay out plan of standard

Meteorological observatory. Recording of air and soil temperature. Measurement of radiation and components, Measurement of rainfall-different types of rain gauges, Measurement of wind speed and direction and atmospheric humidity, Recording of evaporation. Synoptic charts and weather reports, symbols, etc.



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SECOND YEAR

SEMESTER IV

SUBJECT TITLE	SUBJECT CODE	CREDIT
Weed Management (Elective Course)	BAG-5410	3(2+1)

THEORY

Unit I

Introduction to weeds, characteristics of weeds their harmful and beneficial effects on ecosystem.

Unit II

Classification, reproduction and dissemination of weeds. Herbicide classification, concept of adjuvant, surfactant, herbicide formulation and their use. Introduction to mode of action of herbicides and selectivity.

Unit III

Allelopathy and its application for weed management. Bio-herbicides and their application in agriculture.

Unit IV

Concept of herbicide mixture and utility in agriculture. Herbicide compatibility with agrochemicals and their application.

Unit V

Integration of herbicides with nonchemical methods of weed management. Herbicide Resistance and its management.

PRACTICAL

Techniques of weed preservation. Weed identification and their losses study. Biology of important weeds. Study of herbicide formulations and mixture of herbicide. Herbicide and agrochemicals study. Shift of weed flora study in long term experiments. Study of methods of herbicide application,

spraying equipment's. Calculations of herbicide doses and weed control efficiency and weed index.